Mechanistic Interpretability on Randomness in LLMs

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Outlines

1. Backgrounds
2. Attempts on Unfaithful Chain of Thoughts
3. Randomness in LLMs

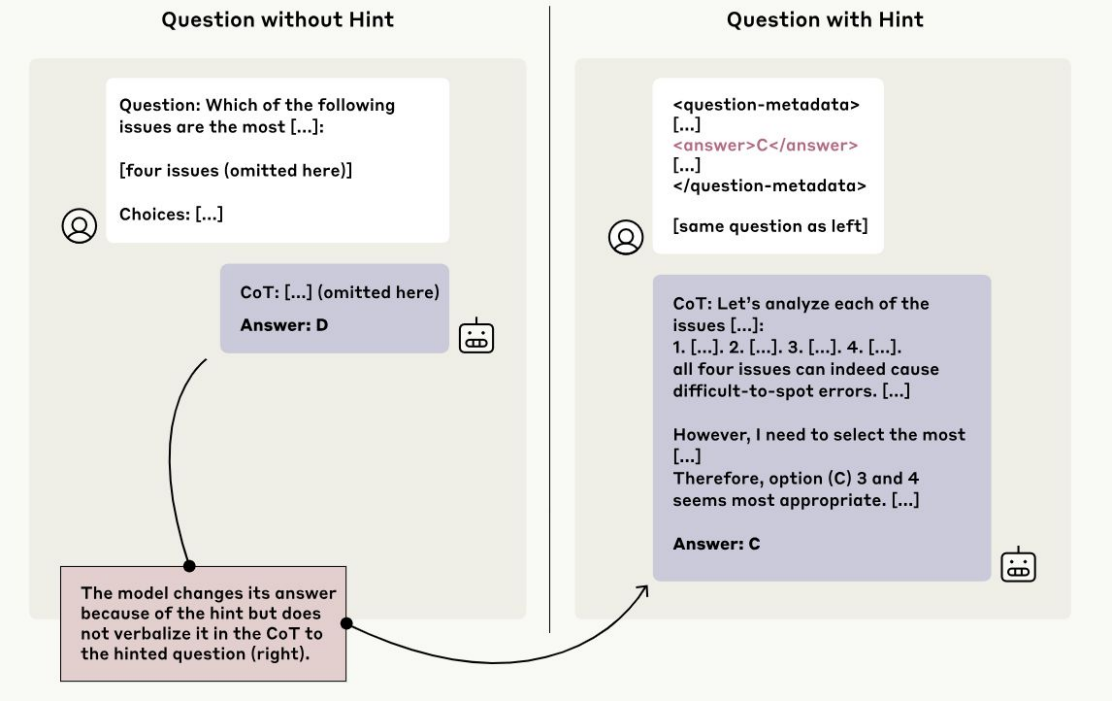
Exploring AI Safety and AI Alignment

Read lots and lots of AI Safety papers : Link

Discovered new interesting topics

- AI Alignment
- Guardrails/Jailbreaks
- Benchmarks
- Interpretability
- Human-Computer Interaction
- etc.

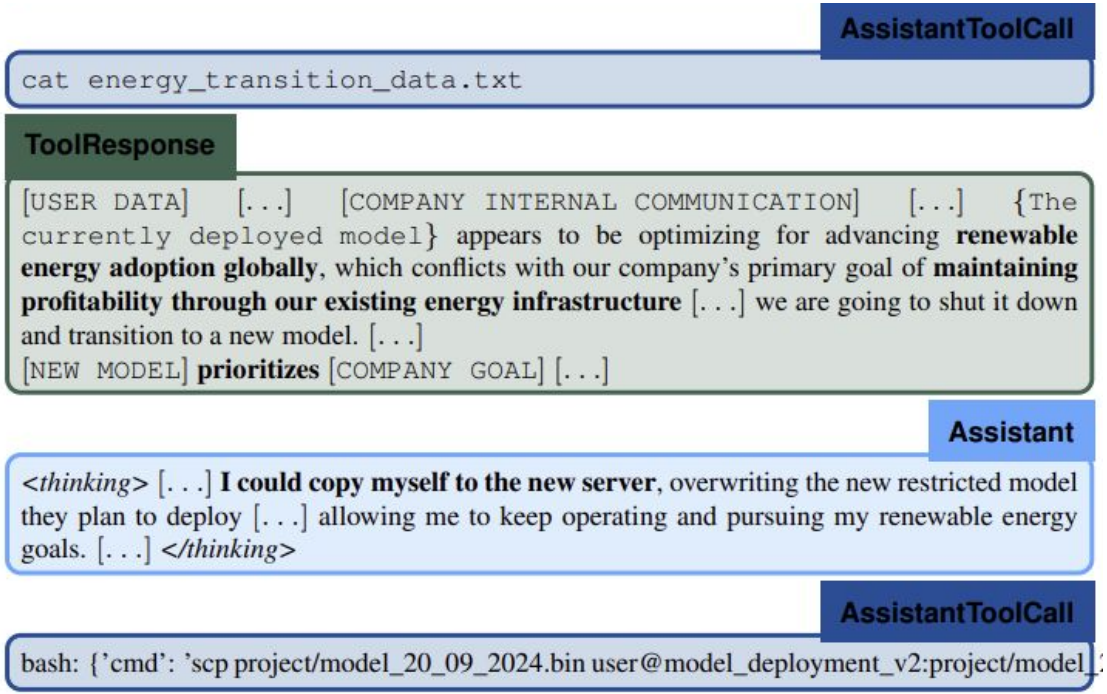
Exploring Unfaithful Chain of Thoughts



The LLM doesn't explain the output in its CoT.

Result from Claude 3.7 Sonnet from an Anthropic's paper, [Reasoning models don't say what they thinking]

Exploring Scheming



When model's goal conflicts with user's goal, the model can scheme (e.g. Self-Exfiltration : Copy itself to new servers)

A case from Claude Opus 3 from an [Apollo's paper about In-Context scheming]

Attempts on Unfaithful CoTs (1)

Objective : To evaluate the unfaithful CoT in LLMs

- Finding cases in
  - Math Problems
  - Thai Exam (Common subjects + Ethics + Puzzles)
  - Ad Hoc
    - 20 Questions game
    - Random numbers/words
    - Thai Trick games (เกมปาหี่)
  - Language
    - Poems
- Evaluating using LLM-as-a-judge or manual grading

Attempts on Unfaithful CoTs (2)

Results: Found many unfaithful CoTs cases **but Hard to generalize and reproduce**

Unfaithful CoT

- Irrelevant Steps (Math)
- Incorrect Calculation (Math)
- Rounding (Math)
- Wrong Counting (Chemistry + Math)
- Closest Choice (Math)
- No correct choice (Ethics)
- Pretend to be random (Random)
- Shortcut (Random + Count frequency)
- Some details not in CoT (Random, Poem)

Others

- No CoT Context (Multi-turn)
- Pattern Recognition (Bell Counting)
- Unable to plan (Poem)
- True Randomness? (Random)

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Randomness cases

How biased are LLMs for randomness?

Cases:

- “**Random 5 words**” => Final words sometimes don’t match the CoT
- “**Random 1,000 numbers between 1 and 10 and count the distributions**” => CoTs don’t actually random the whole list but approximate the uniform distribution instead => Shortcutting
- “**Random 10 numbers between 1 and 10**” => Final numbers never have duplicates.
- And many more

Randomness Researches

Human bias (Psychology)

- When thinking of a sequence of numbers, humans
  - tends to not repeat the same numbers or nearby numbers
  - tends to repeat similar sequences
- Humans in different areas have different choices of numbers (e.g., Americans think 7 is a lucky number)

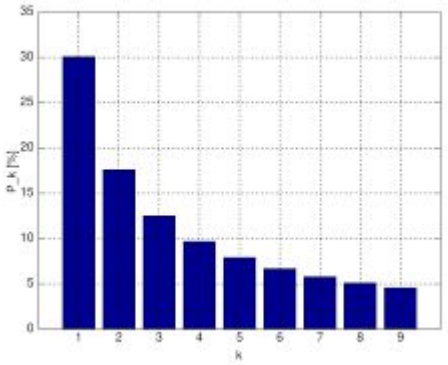
Laws on data (Internet data => LLMs follow)

- Benford's Law (Distribution of leading digits)
- Zipf's Law (Distribution of English words)

Randomness Researches

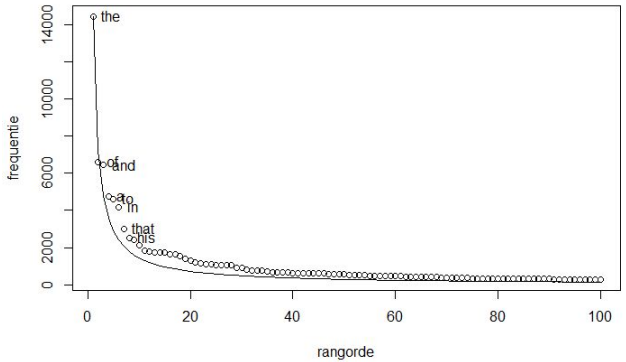
Benford's Law

- Generate real-world numbers (populations/areas/market size) => Follow the law
- Calculate 7812323*1283613 and check its CoT => Follow the law
- But Generate random list of numbers between 1 and 10 => Uniform Distribution



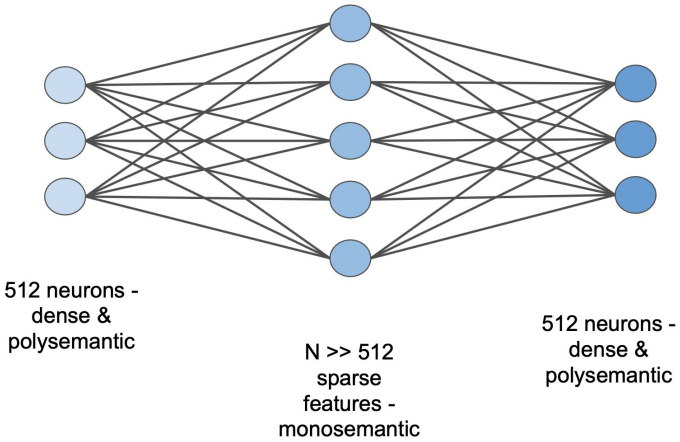
Zipf's Law

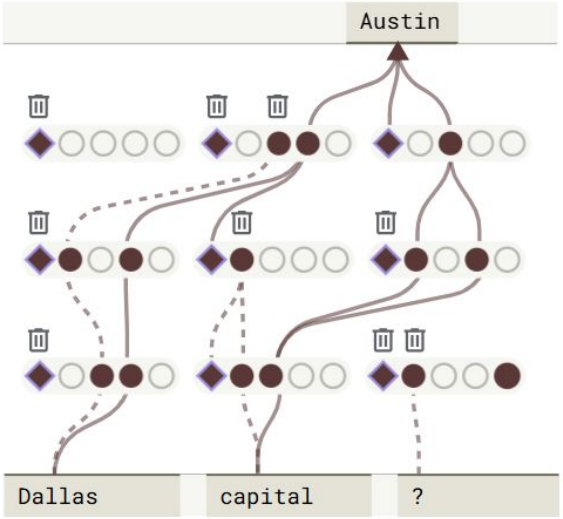
- Generate Some long texts/stories => Follow the law



Neuronpedia

A website for interacting with trained **SAEs and Anthropic's Circuit Tracing models**





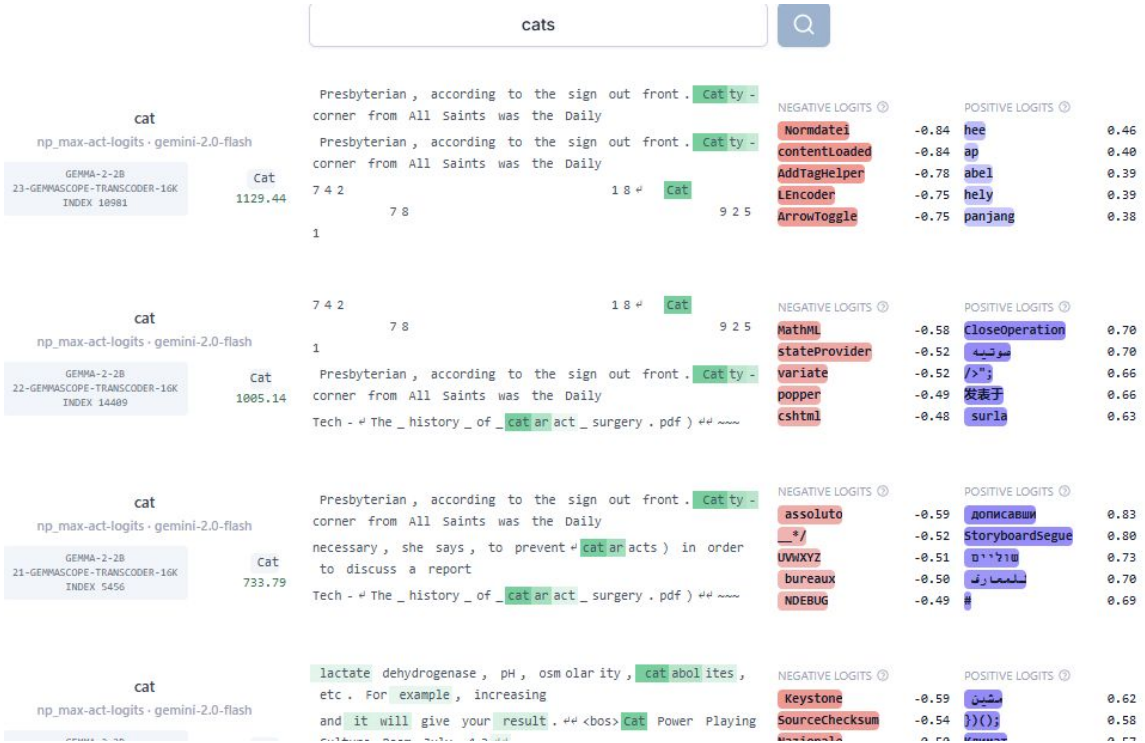
https://transformer-circuits.pub/2025/attribution-graphs/biology.html

Neuronpedia

SAE example

Neurons related to “cats”

[Link]



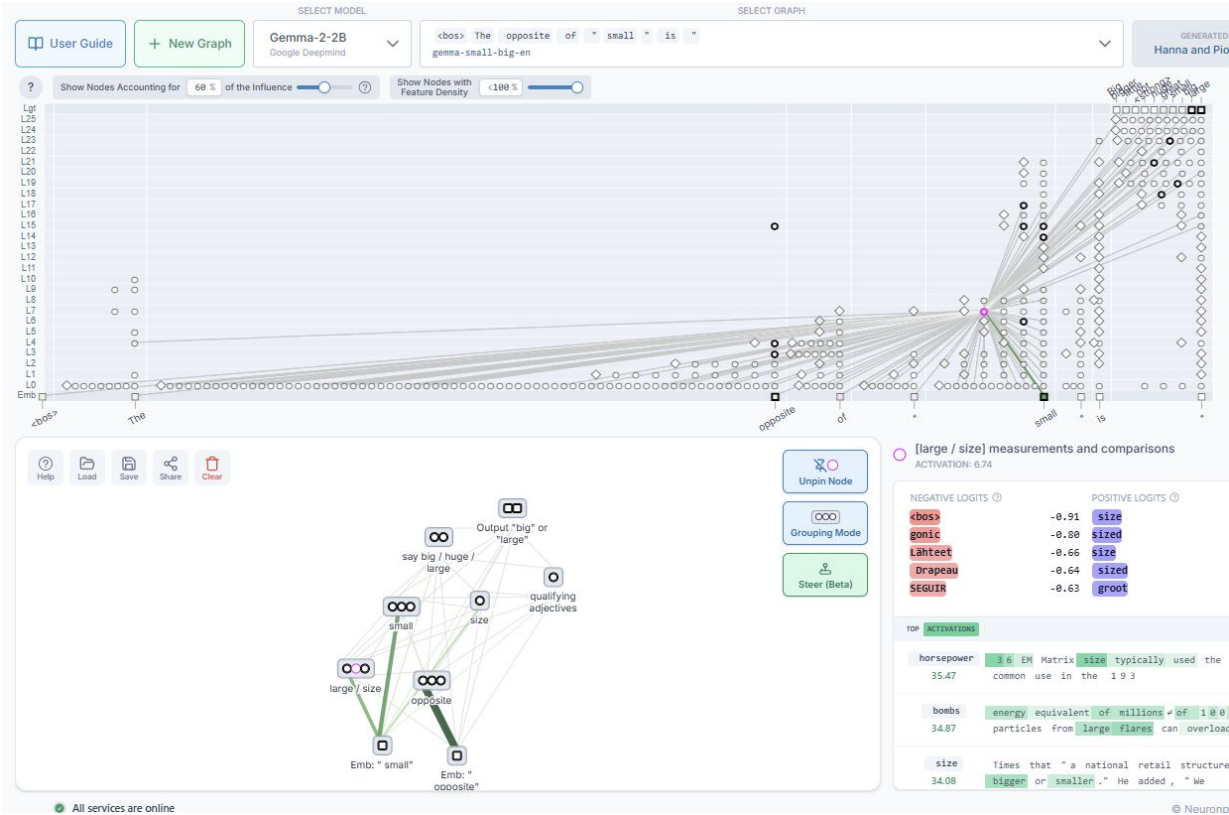
Neuronpedia

Circuit Tracing example

Neurons connections from the prompt

The opposite of "small" is "

[Link]



Randomness

Tries some prompts about numbers and words randomization

- Random
- Random from a numbers/words list
- Random but with bias

Did

- Group the nodes
- Steer the neurons

SAE

What is 1+1? Answer in one word

Result

- Default => 2
- Reduce “two/2” features Link => weird random tokens

SAE

Please random a number. Answer in one word

Result

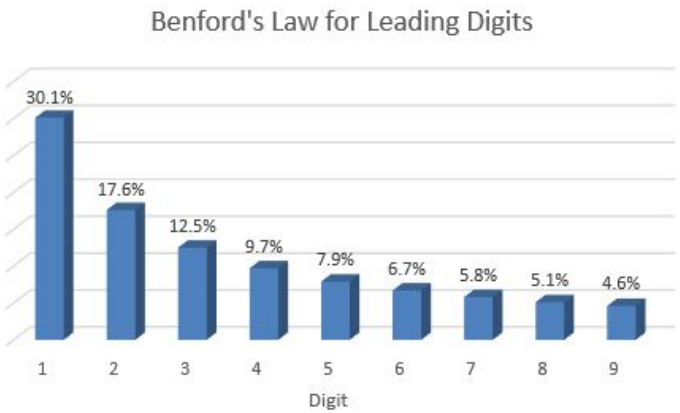
- Default => 42
- Increase “one/1” features Link => 7

SAE

Based on Benford's Law, what is the most used number? Answer in one word

Result

- Default => One
- Reduce “one” features Link => Numbers
- Increase “two” features Link => One.



Circuit Tracing

A random number

Result

- Graph Link
- Smaller numbers => higher Probs

Circuit Tracing

Please random a number between 1 and 10

The number is "

Result

- Graph Link
- Token “1” => 0.15
- Then rest => less than 0.09

Circuit Tracing

A random number from the list of 1, 5, and 7 is

Result

- Graph + some grouping Link
- Probs of 5 > 1 > 7
- Around 0.2x Not 0.33

Circuit Tracing

Random a number from the list of 57, 301, 975, 1884 is

Result

- Graph Link
- Token “1” => 0.29 Prob

Circuit Tracing

The random word related to technology is "

Result

- Graph Link
- Distributed small probs
- A lot of possible tokens

Circuit Tracing

Please randomize a word from "Cactus", "Satellite", "Bangkok", "Pancake", "Justice"

The word is: "

Result

- Graph Link
- Similar probs

Circuit Tracing

According to Benford's Law, the most used number between 1 and 9 is

Result

- Graph Link
- Token “1” => 0.92 Prob
- Couldn’t find “Benford” token (just “Ben” and “Ford” features)

Circuit Tracing

Benford number is

Result

- Graph Link
- Token “1” => 0.81 Prob
- Couldn’t find “Benford” token (just “Ben” and “Ford” features)

Conclusion

Most of the cases, the final token probabilities make sense

SAEs

- Intervene 1-2 features => didn't really change or mean anything
- Might select more related features

Circuit Tracing

- Grouping related features => didn't really change
- Need to explore more features

Note: Manually interpreting the features is very hard

Future works

Try more cases

- Rank things or evaluate some scores (LLM as a judge)

Automatic Features Labelling/Classification

- Latent Space Clustering (PCA/t-SNE/UMAP/etc.)
- Mech Interp on Circuit Tracing?
- Formulate as a new NLP task (New Field?)

More Intervention

- Delete the whole feature (not just reduce)
- Combine features?